



# Outline

## 1 Introduction

- Motivation
- Infrared Triangle
- Soft Theorems
- Asymptotic Symmetries
- Scalar Fields

## 2 Work Plan

- Facts
- Hypothesis
- Objectives

## 3 Methodology

## 4 Work planning

What physical information is encoded at infinity?

# Introduction

What physical information is encoded at infinity?

We study the asymptotic structure of massless scalar fields in flat spacetime. Why?

- Unexpectedly instructive arena!

We will focus in the consequences of relaxing fall of conditions.

**Final goal of this proposal:** to determine how general (relaxed) the asymptotic expansion of a scalar field can be.



# Motivation: Asymptotic symmetries

The study of asymptotic structure in gravity has gained attention from AdS/CFT conjecture [[Maldacena 98'](#)].

An example in  $\text{AdS}_3$  [[Brown-Henneaux 86'](#)]

# Motivation: Asymptotic symmetries

The study of asymptotic structure in gravity has gained attention from AdS/CFT conjecture [Maldacena 98'].

An example in  $\text{AdS}_3$  [Brown-Henneaux 86']

## Asymptotic symmetries of $\text{AdS}_3 \rightarrow$ Conformal Symmetry in 2D

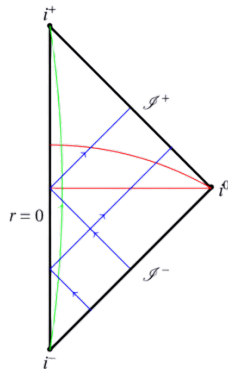
It revealed a nontrivial asymptotic symmetry group acting on the spacetime boundary:

$$[L_m^\pm, L_n^\pm] = (m - n)L_{m+n}^\pm + \frac{c}{12}m(m^2 - 1)\delta_{m+n,0}$$

$$c = \frac{3\ell}{2G} : \quad \text{Brown-Henneaux central charge}$$

# Motivation: Infinities

- Spatial infinity  $i^0$
- Future (past) timelike infinity  $i^+$  ( $i^-$ )
- Future (past) null infinity  $\mathcal{I}^+$  ( $\mathcal{I}^-$ )



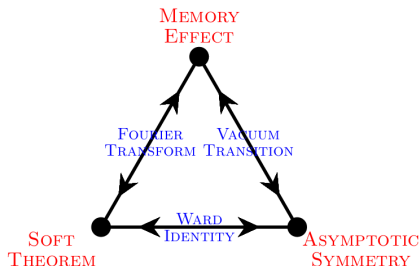
# Motivation: Infrared triangle

Various novel results that give us a new interest in having a deep understanding of the asymptotic structure of a physical theory.

# Motivation: Infrared triangle

Various novel results that give us a new interest in having a deep understanding of the asymptotic structure of a physical theory.

Example: **The infrared triangle** [Strominger 2017]



# Motivation: Infrared Triangle

## Soft theorems

Characterize universal properties of scattering amplitudes when a massless particle in the process becomes soft (limit of zero energy).

# Motivation: Infrared Triangle

## Soft theorems

Characterize universal properties of scattering amplitudes when a massless particle in the process becomes soft (limit of zero energy).

## Asymptotic symmetries

Symmetry transformations that preserve the asymptotic behavior when acting on a field.

# Motivation: Infrared Triangle

## Soft theorems

Characterize universal properties of scattering amplitudes when a massless particle in the process becomes soft (limit of zero energy).

## Asymptotic symmetries

Symmetry transformations that preserve the asymptotic behavior when acting on a field.

## Memory effect

Corresponds to a shift in the relative positions of two inertial detectors when a gravitational wave interact with them.



# What is a soft theorem?

The diagram illustrates a soft theorem. On the left, a shaded circular vertex is connected to several external lines. Two lines are labeled  $p_i$  and  $q$ , where  $q$  is represented by a wavy line. This is followed by an equals sign and a summation symbol  $\sum_{i \in \text{ext}}$ . To the right of the summation is another Feynman diagram where the external line  $q$  is now labeled  $p_i + q$ , and the vertex is connected to the same set of external lines as before. This is followed by a plus sign and an ellipsis  $+$   $\dots$ .

The tree level amplitude factorizes as [\[Weinberg 65' / Cachazo-Strominger 2014\]](#)

$$\lim_{\omega \rightarrow 0} \mathcal{A}_{n+1} = [\omega^{-1} S_n^{(0)} + \omega^0 S_n^{(1)}] \mathcal{A}_n + \mathcal{O}(\omega).$$

# What is a soft theorem?

The diagram illustrates a soft theorem at the level of Feynman diagrams. On the left, a central shaded vertex is connected to several external lines. One line is labeled  $p_i$ , and another is labeled  $q$  and is wavy, representing a soft particle. This is equated to a sum over  $i \in \text{ext}$  of a similar diagram where the external line  $q$  is merged with  $p_i$ , labeled  $p_i + q$ , plus other terms indicated by  $+ \dots$ .

The tree level amplitude factorizes as [\[Weinberg 65' / Cachazo-Strominger 2014\]](#)

$$\lim_{\omega \rightarrow 0} \mathcal{A}_{n+1} = [\omega^{-1} S_n^{(0)} + \omega^0 S_n^{(1)}] \mathcal{A}_n + \mathcal{O}(\omega).$$

When the amplitude is computed at one-loop generates a logarithmic correction [\[Sahoo-Sen 2018/ Ladda-Sahoo 2018/ Saha-Sahoo-Sen 2019\]](#)

$$\lim_{\omega \rightarrow 0} \mathcal{A}_{n+1} = \left[ \omega^{-1} S_n^{(0)} - \frac{k^2}{4} \ln \omega S_n^{(ln)} \right] \mathcal{A}_n + \mathcal{O}(\omega^{(0)}).$$

# Asymptotic Symmetries

The asymptotic symmetries has been attracted people from a several years. Some examples are

- AdS gravity in  $3D \longrightarrow$  Conformal symmetry in 2 dimensions
- General Relativity in  $4D \longrightarrow$  Infinite- dimensional BMS group acting on the boundary.

# IR Triangle Extensions

- Higher dimensions [Kapec-Strominger-Vyacheslav 2014].
- Subleading soft theorems [Campiglia-Laddha 2014, Lysov-Pasterski-Strominger 2014].
- Massive soft particles [Campiglia-Laddha 2015].
- Spin  $\frac{1}{2}$  and spin  $\frac{3}{2}$  soft particles [Avery-Schwab 2015].
- Higher spins [Campoleoni-Francia-Heissenberg 2017].

# The scalar field

The scalar field has been studied in recent works [[Campiglia-Coito-Mizera 2018](#), [Henneaux-Troessaert 2018](#), [Fuentealba-Henneaux 2024](#)].

- Free massless scalar field in Minkowski spacetime

$$S = -\frac{1}{2} \int d^4x \partial_\mu \phi \partial^\mu \phi$$

$$\square \phi = 0$$

- Simplest model to explore infrared structure.

# Asymptotic expansion and conserved charges

This scalar field is considered to be massless, so it is assumed that at spatial infinity it behaves as

$$\phi(t, r, x^A) = \frac{\bar{\phi}(t, x^A)}{r} + \frac{\phi^{(1)}(t, x^A)}{r^2} + \mathcal{O}(r^{-3})$$

It was shown that the leading term defines a conserved quantity

$$\partial_0 \bar{\phi} = 0$$

# Asymptotic expansion and conserved charges

This scalar field is considered to be massless, so it is assumed that at spatial infinity it behaves as

$$\phi(t, r, x^A) = \frac{\bar{\phi}(t, x^A)}{r} + \frac{\phi^{(1)}(t, x^A)}{r^2} + \mathcal{O}(r^{-3})$$

It was shown that the leading term defines a conserved quantity

$$\partial_0 \bar{\phi} = 0$$

In the free case,

$$\pi(t, r, x^A) = \bar{\pi}(t, x^A) + \frac{\pi^{(1)}(t, x^A)}{r} + \mathcal{O}(r^{-2})$$

# Asymptotic expansion and conserved charges

This scalar field is considered to be massless, so it is assumed that at spatial infinity it behaves as

$$\phi(t, r, x^A) = \frac{\bar{\phi}(t, x^A)}{r} + \frac{\phi^{(1)}(t, x^A)}{r^2} + \mathcal{O}(r^{-3})$$

It was shown that the leading term defines a conserved quantity

$$\partial_0 \bar{\phi} = 0$$

In the free case,

$$\pi(t, r, x^A) = \bar{\pi}(t, x^A) + \frac{\pi^{(1)}(t, x^A)}{r} + \mathcal{O}(r^{-2})$$

there is also another conserved quantity

$$\partial_t^2 \phi^{(2)} = 0 \longleftrightarrow \partial_t \bar{\pi} = 0$$



# Outline

## 1 Introduction

- Motivation
- Infrared Triangle
- Soft Theorems
- Asymptotic Symmetries
- Scalar Fields

## 2 Work Plan

- Facts
- Hypothesis
- Objectives

## 3 Methodology

## 4 Work planning

# Hypothesis of the problem: Facts

**F1.** New logarithmic soft theorems by [\[Sahoo-Sen 2018/ Boschetti-Campiglia 2026\]](#).

$$S = (\omega^{-1} S^{(0)} + \ln \omega S^{\text{ln}} + S^{(1)} \omega^0 + \mathcal{O}(\omega \ln \omega)).$$

# Hypothesis of the problem: Facts

**F1.** New logarithmic soft theorems by [\[Sahoo-Sen 2018/ Boschetti-Campiglia 2026\]](#).

$$S = (\omega^{-1} S^{(0)} + \ln \omega S^{\ln} + S^{(1)} \omega^0 + \mathcal{O}(\omega \ln \omega)).$$

**F2.** Finite flux energy with a logarithmically relaxed fall-off in retarded time (log) at  $\mathcal{I}$  [\[Briceño-Gonzalez-Henneaux-Pérez 2025\]](#).

$$\Delta E = \int_{\mathcal{I}^+} du d^2x \sqrt{g} (\partial_u a)^2 \Leftrightarrow \phi \sim \frac{1}{r} a(u, x^A) \sim \frac{\ln u}{r} \phi^{(1)}(\bar{x}^A)$$

But  $\phi \sim 1/r$  from a Cauchy surface with appropriate boundary conditions at spatial infinity [\[Henneaux-Troessaert 2018/ Fuentealba-Henneaux 2024\]](#).

# Hypothesis of the problem

The principal ideas of this proposal can be formulated in the following way

- H1.** Slower fall-off of the scalar field at  $i_0$ ? (without comprising invariance and finiteness)

$$\phi \sim r^{-\Delta} \quad \Delta < 1 \quad (\text{compact support?})$$

# Hypothesis of the problem

The principal ideas of this proposal can be formulated in the following way

- H1.** Slower fall-off of the scalar field at  $i_0$ ? (without comprising invariance and finiteness)

$$\phi \sim r^{-\Delta} \quad \Delta < 1 \quad (\text{compact support?})$$

- H2.** New conserved charges associated to this presumed asymptotic behavior?

# Hypothesis of the problem

The principal ideas of this proposal can be formulated in the following way

- H1.** Slower fall-off of the scalar field at  $i_0$ ? (without comprising invariance and finiteness)

$$\phi \sim r^{-\Delta} \quad \Delta < 1 \quad (\text{compact support?})$$

- H2.** New conserved charges associated to this presumed asymptotic behavior?

- H3.** Controlled identification of possible decays (physically admissible) that can be extended to gauge and gravitational theories.

# Objectives

The principal objectives of this work is to investigate the relaxation of the fall-off conditions of massless scalar fields.

# Objectives

The principal objectives of this work is to investigate the relaxation of the fall-off conditions of massless scalar fields.

- OE1.** Propose and classify families of relaxed fall-off conditions for the scalar field.



# Objectives

The principal objectives of this work is to investigate the relaxation of the fall-off conditions of massless scalar fields.

- OE1.** Propose and classify families of relaxed fall-off conditions for the scalar field.
- OE2.** Analyze the variational consistency of the proposed boundary conditions and determine the criteria for the finiteness and integrability of the associated charges.

# Objectives

The principal objectives of this work is to investigate the relaxation of the fall-off conditions of massless scalar fields.

- OE1.** Propose and classify families of relaxed fall-off conditions for the scalar field.
- OE2.** Analyze the variational consistency of the proposed boundary conditions and determine the criteria for the finiteness and integrability of the associated charges.
- OE3.** Calculate the conserved charges and matching conditions for each class of fall-off considered.
- OE4.** Extend the new fall-off to the dual formulation of the scalar field as a gauge 2-form Kalb-Ramond field.

# Outline

## 1 Introduction

- Motivation
- Infrared Triangle
- Soft Theorems
- Asymptotic Symmetries
- Scalar Fields

## 2 Work Plan

- Facts
- Hypothesis
- Objectives

## 3 Methodology

## 4 Work planning

# Methodology

We first study the symmetry of Minkowski spacetime. In spherical coordinates

# Methodology

We first study the symmetry of Minkowski spacetime. In spherical coordinates

$$ds^2 = -dt^2 + dr^2 + r^2 \bar{g}_{AB} d\bar{x}^A d\bar{x}^B$$

The Killing vectors are obtained via Lie derivative

$$\mathcal{L}_\xi g_{\mu\nu} = \xi^\lambda \partial_\lambda g_{\mu\nu} + \partial_\mu \xi^\lambda g_{\lambda\nu} + \partial_\nu \xi^\lambda g_{\mu\lambda}$$

# Methodology

We first study the symmetry of Minkowski spacetime. In spherical coordinates

$$ds^2 = -dt^2 + dr^2 + r^2 \bar{g}_{AB} d\bar{x}^A d\bar{x}^B$$

The Killing vectors are obtained via Lie derivative

$$\mathcal{L}_\xi g_{\mu\nu} = \xi^\lambda \partial_\lambda g_{\mu\nu} + \partial_\mu \xi^\lambda g_{\lambda\nu} + \partial_\nu \xi^\lambda g_{\mu\lambda}$$

and they are

$$\xi^t = br + T_0$$

$$\xi^r = W$$

$$\xi^A = \bar{Y}^A + \frac{1}{r} \bar{D}^A W$$

They correspond to Lorentz boosts ( $b$ ), spatial rotations ( $Y^A$ ) and space and time translations ( $W$  and  $T_0$ ).

As a first step, we will analyze what fall-off is admissible. For example:

$$\phi(r, t, \bar{x}^A) = \frac{\log r}{r} \phi_{\log}(t, \bar{x}^A) + \frac{\bar{\phi}(t, \bar{x}^A)}{r} + \frac{\phi^{(1)}(t, \bar{x}^A)}{r^2} + o(r^{-2})$$

$$\pi(r, t, \bar{x}^A) = \log r \pi_{\log}(t, \bar{x}^A) + \bar{\pi}(t, \bar{x}^A) + \frac{\pi^{(1)}(t, \bar{x}^A)}{r} + o(r^{-1}).$$

We need to check if this is a **physically admissible** relaxed fall-off.

To check what configurations are admissible, we check the following properties

- Fall-off invariance under Poincaré group.
- Finiteness of the symplectic form  $\Omega$ .
- Invariance over Poincaré  $i_\xi \Omega = 0$ .
- Finiteness of the canonical generator  $i_\xi \Omega = -d_V Q_\xi$ .



# Methodology

Analysis is carried out in the Hamiltonian prescription of the field theory (symplectic structures).

# Methodology

Analysis is carried out in the Hamiltonian prescription of the field theory (symplectic structures).

In order to check the fall-off invariance, we compute the transformation laws of the scalar field under Poincaré

$$\begin{aligned}\delta_\xi \phi &= \frac{\xi^\perp \pi}{\sqrt{\gamma}} + \xi^i \partial_i \phi \\ \delta_\xi \pi &= \partial_i (\sqrt{\gamma} \xi^\perp \gamma^{ij} \partial_j \phi) + \partial_i (\xi^i \pi)\end{aligned}$$

# Methodology

Analysis is carried out in the Hamiltonian prescription of the field theory (symplectic structures).

In order to check the fall-off invariance, we compute the transformation laws of the scalar field under Poincaré

$$\begin{aligned}\delta_\xi \phi &= \frac{\xi^\perp \pi}{\sqrt{\gamma}} + \xi^i \partial_i \phi \\ \delta_\xi \pi &= \partial_i (\sqrt{\gamma} \xi^\perp \gamma^{ij} \partial_j \phi) + \partial_i (\xi^i \pi)\end{aligned}$$

After this we analyze if the symplectic form is finite

$$\Omega = \int d^3x d_V (\log r \pi_{\log} + \bar{\pi} + \dots) \wedge d_V \left( \frac{\log r}{r} \phi_{\log} + \frac{\bar{\phi}}{r} + \dots \right).$$

# Methodology

How we deal with the polylogarithmic divergences? **Parity conditions** as boundary conditions:

$$\overline{\phi}, \phi_{\log} \rightarrow \text{even} \quad \overline{\pi}, \pi_{\log} \rightarrow \text{odd}$$

under the antipodal map of the sphere at  $i^0$ .

# Methodology

How we deal with the polylogarithmic divergences? **Parity conditions** as boundary conditions:

$$\overline{\phi}, \phi_{\log} \rightarrow \text{even} \quad \overline{\pi}, \pi_{\log} \rightarrow \text{odd}$$

under the antipodal map of the sphere at  $i^0$ .

Poincaré invariance of the symplectic form  $\Omega$ ? (Cartan's magic formula  $\mathcal{L}_\xi \Omega = \{d_V, i_\xi\} \Omega = 0$ )

$$i_\xi \Omega = \int d^3x (\delta_\xi \pi d_V \phi - d_V \pi \delta_\xi \phi)$$

# Methodology

How we deal with the polylogarithmic divergences? **Parity conditions** as boundary conditions:

$$\overline{\phi}, \phi_{\log} \rightarrow \text{even} \quad \overline{\pi}, \pi_{\log} \rightarrow \text{odd}$$

under the antipodal map of the sphere at  $i^0$ .

Poincaré invariance of the symplectic form  $\Omega$ ? (Cartan's magic formula  $\mathcal{L}_\xi \Omega = \{d_V, i_\xi\} \Omega = 0$ )

$$i_\xi \Omega = \int d^3x (\delta_\xi \pi d_V \phi - d_V \pi \delta_\xi \phi)$$

After the application of the vertical derivative,  $\mathcal{L}_\xi \Omega$  reduces to the following surface integrals

$$\mathcal{L}_\xi \Omega = \int d^2s_i (\sqrt{\gamma} \xi^\perp \gamma^{ij} \partial_j d_V \phi) \wedge d_V \phi + \int d^2s_i \xi^i d_V \pi \wedge d_V \phi$$

Now we can compute the canonical generator by the relation (Poincaré lemma)

$$i_\xi \Omega = -d_V Q$$

From the symplectic form invariant under Poincaré we obtain

$$i_\xi \Omega = -d_V \left[ \int d^3x \left( \xi^\perp \left( \frac{\pi^2}{2\sqrt{\gamma}} + \frac{1}{2} \sqrt{\gamma} \gamma^{ij} \partial_i \phi \partial_j \phi \right) + \xi^i \pi \partial_i \phi \right) \right],$$

from here we can read  $\mathcal{H}_i$  and  $\mathcal{H}_\perp$ . Given the invariance and well-defined symplectic form, this is a canonical generator.

The canonical generator is given by

$$Q_{\xi, \xi^i} = \int d^3x \left( \xi^\perp \mathcal{H}_\perp + \xi^i \mathcal{H}_i \right) + B_{\xi, \xi^i}$$

[Regge-Teitelboim 74']

Are the conservation laws associated to symmetry transformations?



The canonical generator is given by

$$Q_{\xi, \xi^i} = \int d^3x \left( \xi^\perp \mathcal{H}_\perp + \xi^i \mathcal{H}_i \right) + B_{\xi, \xi^i}$$

[Regge-Teitelboim 74']

Are the conservation laws associated to symmetry transformations?

No.

The canonical generator is given by

$$Q_{\xi, \xi^i} = \int d^3x \left( \xi^\perp \mathcal{H}_\perp + \xi^i \mathcal{H}_i \right) + B_{\xi, \xi^i}$$

[Regge-Teitelboim 74']

Are the conservation laws associated to symmetry transformations?

**No.** Dual formulation in terms of the Kalb-Ramond field

$$d\phi = *dB \quad I = \int dB \wedge *dB$$

[Henneaux-Troessaert 2018]

# Methodology

To understand the behavior of initial data at null infinity, we need to solve

$$\square\phi = 0$$

# Methodology

To understand the behavior of initial data at null infinity, we need to solve

$$\square\phi = 0$$

Hyperbolic coordinates helps to resolve spatial infinity.

$$dS^2 = d\eta^2 + \eta^2 h_{ab} dx^a dx^b,$$
$$h_{ab} dx^a dx^b = \frac{-1}{(1-s^2)^2} ds^2 + \frac{1}{1-s^2} \bar{g}_{AB} d\bar{x}^A d\bar{x}^B$$

# Methodology

To understand the behavior of initial data at null infinity, we need to solve

$$\square\phi = 0$$

Hyperbolic coordinates helps to resolve spatial infinity.

$$\begin{aligned} dS^2 &= d\eta^2 + \eta^2 h_{ab} dx^a dx^b, \\ h_{ab} dx^a dx^b &= \frac{-1}{(1-s^2)^2} ds^2 + \frac{1}{1-s^2} \bar{g}_{AB} d\bar{x}^A d\bar{x}^B \end{aligned}$$

In  $s = 0$  we coincide with the hyperplane  $t = 0$  ( $\eta = r$ ). The Klein-Gordon equation becomes

$$\mathcal{D}^a \mathcal{D}_a \phi + \eta^{-1} \partial_\eta (\eta^3 \partial_\eta \phi) = 0$$

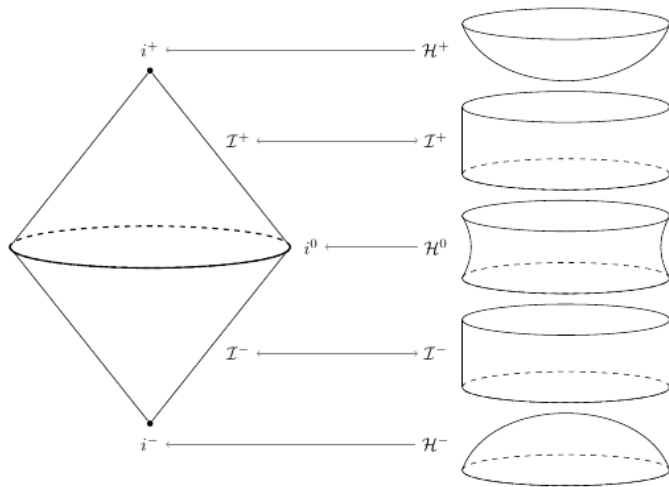


Figure: Boschetti, Campiglia 2026

These coordinates do not touch null infinity. We need to perform the next steps

- Friedrich coordinates.
- Express  $\phi$  in this coordinates.
- Construct conserved quantities at null infinity.
- Connect Spatial and null infinity conserved quantities.
- Matching conditions.

# Outline

## 1 Introduction

- Motivation
- Infrared Triangle
- Soft Theorems
- Asymptotic Symmetries
- Scalar Fields

## 2 Work Plan

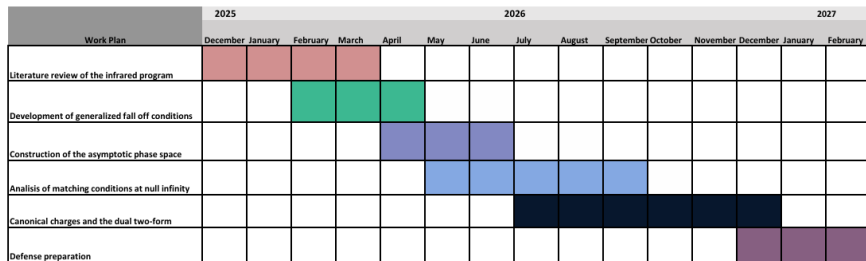
- Facts
- Hypothesis
- Objectives

## 3 Methodology

## 4 Work planning



# Gantt chart



# THANKS FOR YOUR ATTENTION



$$+ \frac{\partial}{\partial x} \frac{\partial^2}{\partial t^2} (v + y_t^2)$$

$$= \frac{\partial^t}{\partial t} \frac{\partial A_\mu}{\partial x} + \frac{1}{2} \sum_{k=1}^{\infty} h_i \left( 1 - F_i \left( i - \frac{1}{k} K_i^{-\frac{1}{2}} \right) \right)$$

$$= \frac{1}{2x} \sum_{k=1}^n \left[ T_i - \eta_i^{27} \right]$$

$$\nabla_\mu A_\nu - \nabla_\nu A_\mu = F_{\mu\nu}$$

$$H = \sum_n E_n |n\rangle \langle n|$$

$$S = \int d^{10}x \sqrt{-g} (R + \dots)$$

Calabi-Yau manifold  
compactification